

R version 4.5.2 (2025-10-31) -- "[Not] Part in a Rumble"
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Platform: aarch64-apple-darwin20

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[R.app GUI 1.82 (8556) aarch64-apple-darwin20]

```
> install.packages(c("lcmm", "ggplot2", "dplyr", "tidyr", "scales"))
--- Please select a CRAN mirror for use in this session ---
also installing the dependencies 'foreach', 'iterators', 'spacefillr', 'marqLevAlg', 'doParallel'

trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/foreach_1.5.2.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/iterators_1.0.14.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/spacefillr_0.4.0.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/marqLevAlg_2.0.8.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/doParallel_1.0.17.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/lcmm_2.2.2.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/ggplot2_4.0.3.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/dplyr_1.2.1.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/tidyr_1.3.2.tgz'
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/scales_1.4.0.tgz'
```

The downloaded binary packages are in
/var/folders/q5/q_63_0fx4l10mbpykrjdd_940000gn/T/RtmpCkGdDh/downloaded_packages
> setwd("/Users/soupornoghosh/Desktop/IMT 600/OpenAlex/FullData")
> source("run_lcga.R")

```
=====
LCGA: I/O Faculty Publication Trajectories
=====
```

Rows loaded: 1777
Faculty loaded: 114
Faculty with ≥ 3 non-missing pub_count observations: 114
Career year range: -2 - 44
Publication count range: 0 - 90 (mean = 4.45)

Using sqrt(pub_count) as the modeled outcome.

```
--- Checking functional form (1-class models) ---
Linear   - BIC: 6019.9  AIC: 6011.6
Quadratic - BIC: 6023.3  AIC: 6012.3
Selected: linear growth function.
```

```
--- Fitting LCGA: 1 to 7 classes (linear growth) ---
```

```

Fitting 1-class model ... BIC = 6019.9
Error in data.frame(n_classes = 1, log_lik = models[[1]]$loglik, AIC = models[[1]]$AIC, :
arguments imply differing number of rows: 1, 0
> source("run_lcga.R")

```

```

=====
LCGA: I/O Faculty Publication Trajectories
=====

```

```

Rows loaded: 1777
Faculty loaded: 114
Dropped 3 rows with career_year < 1 (pre-Asst-Prof observations).
Affected faculty: Preston Lindsay

```

```

Faculty with ≥3 non-missing pub_count observations: 114
Career year range: 1 – 44
Publication count range: 0 – 90 (mean = 4.45)

```

Using sqrt(pub_count) as the modeled outcome.

```

--- Checking functional form (1-class models) ---
Linear   - BIC: 6010.6  AIC: 6002.4
Quadratic - BIC: 6013.7  AIC: 6002.8
Selected: linear growth function.

```

```

--- Fitting LCGA: 1 to 7 classes (linear growth) ---
Fitting 1-class model ... BIC = 6010.6
Fitting 2-class model (B=50 random starts) ... BIC = 5065.7  entropy = 0.936  AvePP = 0.974
Fitting 3-class model (B=50 random starts) ... BIC = 4760.9  entropy = 0.934  AvePP = 0.979
Fitting 4-class model (B=50 random starts) ... BIC = 4523.2  entropy = 0.888  AvePP = 0.937
Fitting 5-class model (B=50 random starts) ... BIC = 4339.7  entropy = 0.923  AvePP = 0.916
Warning: class(es) with < 5 members: 5
Fitting 6-class model (B=50 random starts) ... BIC = 4203.0  entropy = 0.920  AvePP = 0.942
Warning: class(es) with < 5 members: 4, 5
Fitting 7-class model (B=50 random starts) ... BIC = 4194.5  entropy = 0.883  AvePP = 0.907
Warning: class(es) with < 5 members: 5, 6

```

```

--- Model fit comparison ---
n_classes log_lik  AIC  BIC entropy ave_pp n_params
1         -2998 6002 6011      NA      NA         NA
2         -2519 5049 5066 0.9357 0.9743         NA
3         -2359 4736 4761 0.9337 0.9793         NA
4         -2233 4490 4523 0.8881 0.9370         NA
5         -2134 4299 4340 0.9232 0.9161         NA
6         -2059 4154 4203 0.9200 0.9423         NA
7         -2048 4137 4195 0.8832 0.9067         NA

```

```

BIC plateaus at 6 classes (delta < 10).
BIC-optimal solution: 7 classes.

```

```

Fit table saved to 'lcga_fit_comparison.csv'.
BIC elbow plot saved to 'lcga_bic_plot.pdf'.
Class assignments saved to 'lcga_class_assignments.csv'.

```

```

--- Descriptives by class ---

```

```

# A tibble: 7 × 7
  class_assignment n_faculty pct_of_sample mean_career_length mean_pub_per_year
      <int>      <int>      <dbl>      <dbl>      <dbl>
1             1         8          7         12.1         14.2

```

2	2	8	7	17.8	2.61
3	3	51	44.7	13.4	5.46
4	4	15	13.2	17.8	2.56
5	5	2	1.8	35.5	7.07
6	6	2	1.8	14.5	13.4
7	7	28	24.6	17.3	0.25

i 2 more variables: pct_ever_promoted <dbl>, pct_never_asst <dbl>

Descriptives saved to 'lcga_descriptives_by_class.csv'.

Error in x\$call\$fixed[3] : object of type 'symbol' is not subsettable

> source("run_lcga.R")

=====

LCGA: I/O Faculty Publication Trajectories

=====

Rows loaded: 1777

Faculty loaded: 114

Dropped 3 rows with career_year < 1 (pre-Asst-Prof observations).

Affected faculty: Preston Lindsay

Small-class warning threshold: 5 (5% of n=114)

Faculty with ≥3 non-missing pub_count observations: 114

Career year range: 1 – 44

Publication count range: 0 – 90 (mean = 4.45)

Using sqrt(pub_count) as the modeled outcome.

--- Checking functional form (1-class models) ---

Linear – BIC: 6010.6 AIC: 6002.4

Quadratic – BIC: 6013.7 AIC: 6002.8

Selected: linear growth function.

--- Fitting LCGA: 1 to 11 classes (linear growth) ---

Fitting 1-class model ... BIC = 6010.6

Fitting 2-class model (B=50 random starts) ... BIC = 5065.7 entropy = 0.936 AvePP = 0.974

Fitting 3-class model (B=50 random starts) ... BIC = 4760.9 entropy = 0.934 AvePP = 0.979

Fitting 4-class model (B=50 random starts) ... BIC = 4523.2 entropy = 0.888 AvePP = 0.937

Fitting 5-class model (B=50 random starts) ... BIC = 4339.7 entropy = 0.923 AvePP = 0.916

Warning: class(es) with < 5 members: 5

Fitting 6-class model (B=50 random starts) ... BIC = 4203.0 entropy = 0.920 AvePP = 0.942

Warning: class(es) with < 5 members: 4, 5

Fitting 7-class model (B=50 random starts) ... BIC = 4194.5 entropy = 0.883 AvePP = 0.907

Warning: class(es) with < 5 members: 5, 6

Fitting 8-class model (B=50 random starts) ... BIC = 4187.6 entropy = 0.858 AvePP = 0.885

Warning: class(es) with < 5 members: 4, 6

Fitting 9-class model (B=50 random starts) ... BIC = 4184.8 entropy = 0.859 AvePP = 0.872

Warning: class(es) with < 5 members: 1, 2, 6

Fitting 10-class model (B=50 random starts) ... BIC = 4184.6 entropy = 0.825 AvePP = 0.856

Warning: class(es) with < 5 members: 3, 7, 8

Fitting 11-class model (B=50 random starts) ... BIC = 4192.7 entropy = 0.776 AvePP = 0.825

Warning: class(es) with < 5 members: 1, 2, 10, 11

--- Model fit comparison ---

n_classes	log_lik	AIC	BIC	entropy	ave_pp	n_params
1	-2998	6002	6011	NA	NA	NA
2	-2519	5049	5066	0.9357	0.9743	NA
3	-2359	4736	4761	0.9337	0.9793	NA
4	-2233	4490	4523	0.8881	0.9370	NA
5	-2134	4299	4340	0.9232	0.9161	NA

6	-2059	4154	4203	0.9200	0.9423	NA
7	-2048	4137	4195	0.8832	0.9067	NA
8	-2037	4122	4188	0.8584	0.8849	NA
9	-2028	4111	4185	0.8594	0.8717	NA
10	-2021	4103	4185	0.8252	0.8563	NA
11	-2018	4102	4193	0.7761	0.8248	NA

BIC plateaus at 6 classes (delta < 10).
 BIC-optimal solution: 10 classes.

Fit table saved to 'lcga_fit_comparison.csv'.
 BIC elbow plot saved to 'lcga_bic_plot.pdf'.
 Class assignments saved to 'lcga_class_assignments.csv'.

--- Descriptives by class ---

A tibble: 10 × 7

	class_assignment	n_faculty	pct_of_sample	mean_career_length	mean_pub_per_year
	<int>	<int>	<dbl>	<dbl>	<dbl>
1	1	7	6.1	16	2.32
2	2	5	4.4	10.2	16.1
3	3	2	1.8	14.5	13.4
4	4	49	43	13.2	5.51
5	5	7	6.1	20.7	0.76
6	6	12	10.5	17.4	2.34
7	7	3	2.6	15.3	10.9
8	8	2	1.8	35.5	7.07
9	9	6	5.3	20.8	3.9
10	10	21	18.4	16.2	0.08

i 2 more variables: pct_ever_promoted <dbl>, pct_never_asst <dbl>
 Descriptives saved to 'lcga_descriptives_by_class.csv'.

Model parameters (best model):

	parameter	estimate
intercept	class1	-0.9273
intercept	class2	-1.4249
intercept	class3	-2.2981
intercept	class4	0.7832
intercept	class5	-0.7967
intercept	class6	-0.5377
intercept	class7	-1.7040
intercept	class8	-2.0412
intercept	class9	-0.5789
intercept	class1	1.4397
intercept	class2	4.1153
intercept	class3	-0.8939
intercept	class4	2.1410
intercept	class5	0.2644
intercept	class6	0.9884
intercept	class7	3.1109
intercept	class8	-0.3688
intercept	class9	1.8846
intercept	class10	0.0648
career_year	class1	-0.0091
career_year	class2	-0.0232
career_year	class3	0.4217
career_year	class4	0.0189
career_year	class5	0.0200
career_year	class6	0.0396

```

career_year class7 0.0014
career_year class8 0.1316
career_year class9 -0.0049
career_year class10 0.0023
stderr 0.6867

```

Trajectory plots saved to 'lcga_trajectories.pdf'.
Model summary saved to 'lcga_model_summary.txt'.

=====

LCGA complete.

Selected model: 10 classes (linear growth)
BIC = 4184.6 Entropy = 0.825

=====

Warning messages:

```

1: Removed 261 rows containing missing values or values outside the scale range
(`geom_line()`).
2: Removed 96 rows containing missing values or values outside the scale range
(`geom_point()`).
3: Removed 261 rows containing missing values or values outside the scale range
(`geom_line()`).
> source("compare_lcga_solutions.R")

```

=====

Comparing LCGA Solutions: 4, 5, and 6 Classes

=====

Faculty in analysis: 114
Career year range: 1 – 44
Plot y-axis cap: 15 publications (95th pct of observed data)

Fitting 1-class base model...
BIC = 6010.6

=====

SOLUTION: 4 CLASSES

=====

Fitting 4-class model (50 random starts)...
BIC = 4523.2 | Entropy = 0.888 | AvePP = 0.937
All classes meet the 5% threshold.
Assignments saved: lcga_4class_assignments.csv

Descriptives (4 classes):

```

# A tibble: 4 × 12
  class_assignment n_faculty pct_of_sample meets_5pct_threshold mean_career_length
      <int>      <int>      <dbl> <lgl>                <dbl>
1         1        28        24.6 TRUE                17.3
2         2        55        48.2 TRUE                14.4
3         3         6         5.3 TRUE                 9.2
4         4        25        21.9 TRUE                17.7
# i 7 more variables: sd_career_length <dbl>, mean_pub_per_year <dbl>,
#   sd_pub_per_year <dbl>, median_pub_per_year <dbl>, pct_ever_promoted <dbl>,
#   pct_multi_promoted <dbl>, pct_never_asst_prof <dbl>
Descriptives saved: lcga_4class_descriptives.csv

```

Model parameters (4 classes):
Heterogenous linear mixed model
fitted by maximum likelihood method

hlme(fixed = pub_sqrt ~ career_year, mixture = ~career_year,

```
subject = "faculty_id", ng = k, data = df, verbose = FALSE)
```

Statistical Model:

```
Dataset: df
Number of subjects: 114
Number of observations: 1774
Number of latent classes: 4
Number of parameters: 12
```

Iteration process:

```
Convergence criteria satisfied
Number of iterations: 1
Convergence criteria: parameters= 5.9e-11
                     : likelihood= 3e-10
                     : second derivatives= 4.9e-11
```

Goodness-of-fit statistics:

```
maximum log-likelihood: -2233.2
AIC: 4490.41
BIC: 4523.24
```

Maximum Likelihood Estimates:

Fixed effects in the class-membership model:
(the class of reference is the last class)

	coef	Se	Wald	p-value
intercept class1	0.10803	0.29964	0.361	0.71846
intercept class2	0.69370	0.27320	2.539	0.01111
intercept class3	-1.35179	0.48074	-2.812	0.00492

Fixed effects in the longitudinal model:

	coef	Se	Wald	p-value
intercept class1	0.13371	0.06709	1.993	0.04628
intercept class2	2.07141	0.05067	40.879	0.00000
intercept class3	2.12197	0.18330	11.577	0.00000
intercept class4	1.14952	0.08078	14.230	0.00000
career_year class1	0.00738	0.00443	1.666	0.09578
career_year class2	0.03073	0.00333	9.236	0.00000
career_year class3	0.24223	0.01797	13.476	0.00000
career_year class4	0.02974	0.00409	7.268	0.00000

	coef	Se
Residual standard error:	0.79619	0.01349

```
Error in summary(model)$coef : $ operator is invalid for atomic vectors
> source("compare_lcgasolutions.R")
```

```
=====
Comparing LCGA Solutions: 4, 5, and 6 Classes
=====
```

```
Faculty in analysis: 114
Career year range: 1 - 44
Plot y-axis cap: 15 publications (95th pct of observed data)
```

Fitting 1-class base model...

BIC = 6010.6

=====

SOLUTION: 4 CLASSES

=====

Fitting 4-class model (50 random starts)...

BIC = 4523.2 | Entropy = 0.888 | AvePP = 0.937

All classes meet the 5% threshold.

Assignments saved: lcga_4class_assignments.csv

Descriptives (4 classes):

A tibble: 4 × 12

	class_assignment	n_faculty	pct_of_sample	meets_5pct_threshold	mean_career_length
	<int>	<int>	<dbl>	<lgl>	<dbl>
1	1	28	24.6	TRUE	17.3
2	2	55	48.2	TRUE	14.4
3	3	6	5.3	TRUE	9.2
4	4	25	21.9	TRUE	17.7

i 7 more variables: sd_career_length <dbl>, mean_pub_per_year <dbl>,

sd_pub_per_year <dbl>, median_pub_per_year <dbl>, pct_ever_promoted <dbl>,

pct_multi_promoted <dbl>, pct_never_asst_prof <dbl>

Descriptives saved: lcga_4class_descriptives.csv

Model parameters (4 classes):

	parameter	coef	Se	wald_z	p_value	sig	count_scale_approx
	intercept class1	0.13371	0.00063	211.690	0	***	0.02
	intercept class2	2.07141	0.00000	Inf	0	***	4.29
	intercept class3	2.12197	0.00000	Inf	0	***	4.50
	intercept class4	1.14952	0.00781	147.236	0	***	1.32
career_year	class1	0.00738	0.00065	11.415	0	***	NA
career_year	class2	0.03073	0.00000	Inf	0	***	NA
career_year	class3	0.24223	0.00378	64.036	0	***	NA
career_year	class4	0.02974	0.00193	15.430	0	***	NA

Parameters saved: lcga_4class_parameters.csv

Trajectory plots saved: lcga_4class_trajectories.pdf

=====

SOLUTION: 5 CLASSES

=====

Fitting 5-class model (50 random starts)...

BIC = 4339.7 | Entropy = 0.923 | AvePP = 0.916

Warning: class(es) below 5% threshold (n<5): Class 5 (n=2)

Assignments saved: lcga_5class_assignments.csv

Descriptives (5 classes):

A tibble: 5 × 12

	class_assignment	n_faculty	pct_of_sample	meets_5pct_threshold	mean_career_length
	<int>	<int>	<dbl>	<lgl>	<dbl>
1	1	52	45.6	TRUE	14
2	2	24	21.1	TRUE	18.2
3	3	28	24.6	TRUE	17.3
4	4	8	7	TRUE	12.1
5	5	2	1.8	FALSE	14.5

i 7 more variables: sd_career_length <dbl>, mean_pub_per_year <dbl>,

sd_pub_per_year <dbl>, median_pub_per_year <dbl>, pct_ever_promoted <dbl>,

pct_multi_promoted <dbl>, pct_never_asst_prof <dbl>

Descriptives saved: lcga_5class_descriptives.csv

Model parameters (5 classes):

	parameter	coef	Se	wald_z	p_value	sig	count_scale_approx
	intercept class1	1.95484	0.00312	625.986	0.00000	***	3.82
	intercept class2	1.11670	0.00385	290.289	0.00000	***	1.25
	intercept class3	0.12572	0.00000	Inf	0.00000	***	0.02
	intercept class4	3.72026	0.00024	15254.621	0.00000	***	13.84
	intercept class5	-0.92052	0.06639	-13.864	0.00000	***	0.00
career_year	class1	0.03274	0.00000	Inf	0.00000	***	NA
career_year	class2	0.03073	0.01837	1.672	0.09446	.	NA
career_year	class3	0.00773	0.00000	Inf	0.00000	***	NA
career_year	class4	-0.01802	0.00040	-44.920	0.00000	***	NA
career_year	class5	0.42326	0.00061	693.260	0.00000	***	NA

Parameters saved: lcga_5class_parameters.csv

Trajectory plots saved: lcga_5class_trajectories.pdf

=====

SOLUTION: 6 CLASSES

=====

Fitting 6-class model (50 random starts)...

BIC = 4203.0 | Entropy = 0.920 | AvePP = 0.942

Warning: class(es) below 5% threshold (n<5): Class 2 (n=2), Class 5 (n=2)

Assignments saved: lcga_6class_assignments.csv

Descriptives (6 classes):

A tibble: 6 × 12

	class_assignment	n_faculty	pct_of_sample	meets_5pct_threshold	mean_career_length
	<int>	<int>	<dbl>	<lgl>	<dbl>
1	1	8	7	TRUE	12.1
2	2	2	1.8	FALSE	35.5
3	3	24	21.1	TRUE	17.2
4	4	50	43.9	TRUE	13.6
5	5	2	1.8	FALSE	14.5
6	6	28	24.6	TRUE	17.3

i 7 more variables: sd_career_length <dbl>, mean_pub_per_year <dbl>,

sd_pub_per_year <dbl>, median_pub_per_year <dbl>, pct_ever_promoted <dbl>,

pct_multi_promoted <dbl>, pct_never_asst_prof <dbl>

Descriptives saved: lcga_6class_descriptives.csv

Model parameters (6 classes):

	parameter	coef	Se	wald_z	p_value	sig	count_scale_approx
	intercept class1	3.71892	0.00000	Inf	0.00000	***	13.83
	intercept class2	-0.36146	0.00000	-Inf	0.00000	***	0.00
	intercept class3	1.22946	0.00071	1727.917	0.00000	***	1.51
	intercept class4	2.11214	0.00028	7505.481	0.00000	***	4.46
	intercept class5	-0.88023	0.00708	-124.254	0.00000	***	0.00
	intercept class6	0.13220	0.02094	6.312	0.00000	***	0.02
career_year	class1	-0.01819	0.00000	-Inf	0.00000	***	NA
career_year	class2	0.13131	0.00010	1294.367	0.00000	***	NA
career_year	class3	0.02495	0.00000	Inf	0.00000	***	NA
career_year	class4	0.01932	0.00000	Inf	0.00000	***	NA
career_year	class5	0.42079	0.00000	Inf	0.00000	***	NA
career_year	class6	0.00752	0.00205	3.676	0.00024	***	NA

Parameters saved: lcga_6class_parameters.csv

Trajectory plots saved: lcga_6class_trajectories.pdf

=====

Done. Files saved for 4-, 5-, and 6-class solutions.

=====

There were 12 warnings (use warnings() to see them)
> source("sensitivity_lcga.R")

LCGA Sensitivity Checks – 4-Class Models

Faculty: 114 | Career year range: 1–44
Y-axis cap: 15 pubs (95th pct) | Small-class threshold: 5

SPEC: sqrt(pub), Quadratic

Fitting 1-class base model...

Base BIC = 6013.7

Fitting 4-class model (50 random starts)...

BIC = 4516.2 | AIC = 4472.5 | Entropy = 0.905 | AvePP = 0.956

All classes meet the 5% threshold.

Assignments saved: lcga_4class_sqrtquad_assignments.csv

Descriptives (sqrt(pub), Quadratic):

A tibble: 4 × 11

	class_assignment	n_faculty	pct_of_sample	meets_5pct	mean_career_length	sd_career_length
	<int>	<int>	<dbl>	<lgl>	<dbl>	<dbl>
1	1	25	21.9	TRUE	17.7	12.7
2	2	6	5.3	TRUE	9.2	7.6
3	3	28	24.6	TRUE	17.3	10.1
4	4	55	48.2	TRUE	14.4	10.3

i 5 more variables: mean_pub_per_year <dbl>, sd_pub_per_year <dbl>,

median_pub_per_year <dbl>, pct_ever_promoted <dbl>, pct_multi_promoted <dbl>

Descriptives saved: lcga_4class_sqrtquad_descriptives.csv

Parameters (sqrt(pub), Quadratic):

	parameter	coef	Se	wald_z	p_value	sig
	intercept class1	1.04320	0.01095	95.306	0.00000	***
	intercept class2	3.09531	0.00000	Inf	0.00000	***
	intercept class3	0.10032	0.00000	Inf	0.00000	***
	intercept class4	2.08572	0.00000	Inf	0.00000	***
	career_year class1	0.04730	0.00000	Inf	0.00000	***
	career_year class2	-0.06387	0.00000	-Inf	0.00000	***
	career_year class3	0.01347	0.00027	49.929	0.00000	***
	career_year class4	0.02704	0.00000	Inf	0.00000	***
I(career_year^2)	class1	-0.00048	0.00000	-Inf	0.00000	***
I(career_year^2)	class2	0.01439	0.00030	48.584	0.00000	***
I(career_year^2)	class3	-0.00019	0.00000	-Inf	0.00000	***
I(career_year^2)	class4	0.00012	0.00023	0.523	0.60065	

Parameters saved: lcga_4class_sqrtquad_parameters.csv

Trajectories saved: lcga_4class_sqrtquad_trajectories.pdf

SPEC: Raw pub, Linear

Fitting 1-class base model...

Base BIC = 11522.9

Fitting 4-class model (50 random starts)...

BIC = 9994.5 | AIC = 9961.7 | Entropy = 0.898 | AvePP = 0.968

Warning: classes below 5% threshold: Class 2 (n=1)

Assignments saved: lcga_4class_rawlin_assignments.csv

Descriptives (Raw pub, Linear):

```
# A tibble: 4 × 11
  class_assignment n_faculty pct_of_sample meets_5pct mean_career_length sd_career_length
      <int>         <int>         <dbl> <lgl>          <dbl>          <dbl>
1           1           7           6.1 TRUE           10.4           7.8
2           2           1           0.9 FALSE           23            NA
3           3          49          43  TRUE           17.1          11.5
4           4          57          50  TRUE           14.8          10.4
# i 5 more variables: mean_pub_per_year <dbl>, sd_pub_per_year <dbl>,
#   median_pub_per_year <dbl>, pct_ever_promoted <dbl>, pct_multi_promoted <dbl>
Descriptives saved: lcga_4class_rawlin_descriptives.csv
```

Parameters (Raw pub, Linear):

```
      parameter      coef      Se  wald_z p_value sig
intercept class1 14.99706 0.00177 8466.884      0 ***
intercept class2 -19.95258 0.00000      -Inf      0 ***
intercept class3  0.63931 0.00000      Inf      0 ***
intercept class4  3.85147 0.00000      Inf      0 ***
career_year class1 -0.05611 0.00006 -970.169      0 ***
career_year class2  3.80040 0.00413  921.166      0 ***
career_year class3  0.06489 0.00000      Inf      0 ***
career_year class4  0.18526 0.00000      Inf      0 ***
Parameters saved: lcga_4class_rawlin_parameters.csv
Trajectories saved: lcga_4class_rawlin_trajectories.pdf
```

=====

SPEC: Raw pub, Quadratic

=====

Fitting 1-class base model...

Base BIC = 11527.2

Fitting 4-class model (50 random starts)...

BIC = 9792.8 | AIC = 9749.1 | Entropy = 0.903 | AvePP = 0.973

Warning: classes below 5% threshold: Class 2 (n=1)

Assignments saved: lcga_4class_rawquad_assignments.csv

Descriptives (Raw pub, Quadratic):

```
# A tibble: 4 × 11
  class_assignment n_faculty pct_of_sample meets_5pct mean_career_length sd_career_length
      <int>         <int>         <dbl> <lgl>          <dbl>          <dbl>
1           1           6           5.3 TRUE           9.3           8
2           2           1           0.9 FALSE           23            NA
3           3          50          43.9 TRUE           16.8          11.5
4           4          57          50  TRUE           15           10.3
# i 5 more variables: mean_pub_per_year <dbl>, sd_pub_per_year <dbl>,
#   median_pub_per_year <dbl>, pct_ever_promoted <dbl>, pct_multi_promoted <dbl>
Descriptives saved: lcga_4class_rawquad_descriptives.csv
```

Parameters (Raw pub, Quadratic):

```
      parameter      coef      Se  wald_z p_value sig
intercept class1 10.30523 0.00000      Inf 0.00000 ***
intercept class2  4.75737 0.00000      Inf 0.00000 ***
intercept class3  1.18894 0.00037 3185.493 0.00000 ***
intercept class4  4.66916 0.00036 13134.052 0.00000 ***
career_year class1  1.77471 0.00205  865.972 0.00000 ***
career_year class2 -2.13114 0.00000      -Inf 0.00000 ***
career_year class3 -0.03724 0.00004 -923.091 0.00000 ***
career_year class4  0.05713 0.00000      Inf 0.00000 ***
I(career_year^2) class1 -0.08295 0.01550  -5.351 0.00000 ***
```

```

I(career_year^2) class2 0.24720 0.00000      Inf 0.00000 ***
I(career_year^2) class3 0.00300 0.00566      0.531 0.59573
I(career_year^2) class4 0.00372 0.00072      5.142 0.00000 ***
Parameters saved: lcga_4class_rawquad_parameters.csv
Trajectories saved: lcga_4class_rawquad_trajectories.pdf

```

SENSITIVITY COMPARISON TABLE

	label	BIC	AIC	entropy	ave_pp	delta_BIC_vs_main	small_class
	sqrt(pub), Linear [MAIN]	4523.2	4490.4	0.8880	0.9370	0.0	None
	sqrt(pub), Quadratic	4516.2	4472.5	0.9055	0.9556	-7.0	None
	Raw pub, Linear	9994.5	9961.7	0.8979	0.9684	5471.3	Class 2 (n=1)
	Raw pub, Quadratic	9792.8	9749.1	0.9032	0.9732	5269.6	Class 2 (n=1)
	note						
	Primary model						
	Most important check						
	Secondary check						
	Secondary check						

Comparison table saved: lcga_sensitivity_comparison.csv

Sensitivity checks complete.
Files saved: assignments, descriptives, parameters, plots
for sqrtquad, rawlin, rawquad specifications.

There were 24 warnings (use warnings() to see them)
> source("explore_rawquad.R")

Raw pub_count - Quadratic Growth: 1 to 7 Classes

Faculty: 114 | Career years: 1-44
Y-axis cap: 15 (95th pct) | Small-class floor: 5 (5%)

Fitting 1-class base model...
BIC = 11527.2

Fitting 2-class model (50 random starts)...
BIC = 10559.6 | Entropy = 0.994 | AvePP = 0.998 | Small: None
Fitting 3-class model (50 random starts)...
BIC = 10077.8 | Entropy = 0.869 | AvePP = 0.957 | Small: None
Fitting 4-class model (50 random starts)...
BIC = 9792.8 | Entropy = 0.903 | AvePP = 0.973 | Small: C1(n=1) ← SINGLETON CLASS
Fitting 5-class model (50 random starts)...
BIC = 9648.3 | Entropy = 0.919 | AvePP = 0.980 | Small: C4(n=2), C5(n=1) ← SINGLETON CLASS
Fitting 6-class model (50 random starts)...
BIC = 9556.2 | Entropy = 0.844 | AvePP = 0.938 | Small: C4(n=1), C5(n=2) ← SINGLETON CLASS
Fitting 7-class model (50 random starts)...
BIC = 9557.8 | Entropy = 0.837 | AvePP = 0.938 | Small: C1(n=1), C5(n=1), C6(n=1) ← SINGLETON CLASS

--- Fit comparison (raw pub_count, quadratic) ---

n_classes	log_lik	AIC	BIC	entropy	ave_pp	small_class	viable
1	-5754.14	11516.3	11527.2	NA	NA	-	TRUE
2	-5260.86	10537.7	10559.6	0.9937	0.9980	None	TRUE

3	-5010.50	10045.0	10077.8	0.8685	0.9565		None	TRUE
4	-4858.53	9749.1	9792.8	0.9032	0.9732		C1(n=1)	FALSE
5	-4776.78	9593.6	9648.3	0.9192	0.9801		C4(n=2), C5(n=1)	FALSE
6	-4721.26	9490.5	9556.2	0.8437	0.9377		C4(n=1), C5(n=2)	FALSE
7	-4712.58	9481.2	9557.8	0.8365	0.9384	C1(n=1), C5(n=1), C6(n=1)		FALSE

BIC deltas (k vs k-1):

1→2: -967.6
 2→3: -481.8
 3→4: -285.0
 4→5: -144.5
 5→6: -92.1
 6→7: 1.6

BIC-optimal viable solution: 3 classes

BIC plot saved: rawquad_bic_plot.pdf

--- Outputs for 2-class solution ---

Descriptives:

A tibble: 2 × 10

	class_assignment	n_faculty	pct_of_sample	meets_5pct	mean_career_length	mean_pub_per_year
	<int>	<int>	<dbl>	<lgl>	<dbl>	<dbl>
1	1	6	5.3	TRUE	9.2	17.2
2	2	108	94.7	TRUE	15.9	3.66

i 4 more variables: sd_pub_per_year <dbl>, median_pub_per_year <dbl>,

pct_ever_promoted <dbl>, pct_multi_promoted <dbl>

Parameters:

	parameter	coef	Se	wald_z	p_value	sig
	intercept class1	16.13496	0.00000	Inf	0	***
	intercept class2	3.19485	0.00000	Inf	0	***
	career_year class1	-2.34724	0.00000	-Inf	0	***
	career_year class2	0.01475	0.00018	81.983	0	***
I(career_year^2)	class1	0.22698	0.00496	45.730	0	***
I(career_year^2)	class2	0.00248	0.00000	Inf	0	***

Trajectory shapes:

	class	b1	b2	shape	peak_or_trough_year
	Class 1 (n=6)	-2.3472	0.22698	U-shape (dip & rise)	5.2
	Class 2 (n=108)	0.0148	0.00248	U-shape (dip & rise)	-3.0

Trajectories saved: rawquad_2class_trajectories.pdf

--- Outputs for 3-class solution ---

Descriptives:

A tibble: 3 × 10

	class_assignment	n_faculty	pct_of_sample	meets_5pct	mean_career_length	mean_pub_per_year
	<int>	<int>	<dbl>	<lgl>	<dbl>	<dbl>
1	1	53	46.5	TRUE	16.2	1.28
2	2	55	48.2	TRUE	15.6	5.96
3	3	6	5.3	TRUE	9.2	17.2

i 4 more variables: sd_pub_per_year <dbl>, median_pub_per_year <dbl>,

pct_ever_promoted <dbl>, pct_multi_promoted <dbl>

Parameters:

	parameter	coef	Se	wald_z	p_value	sig
	intercept class1	1.26852	0.00000	Inf	0.00000	***
	intercept class2	4.88924	0.00000	Inf	0.00000	***
	intercept class3	16.21482	0.00000	Inf	0.00000	***
	career_year class1	-0.03605	0.00000	-Inf	0.00000	***
	career_year class2	0.08046	0.00149	54.123	0.00000	***

```

      career_year class3 -2.36095 0.00000 -Inf 0.00000 ***
I(career_year^2) class1 0.00292 0.02265 0.129 0.89727
I(career_year^2) class2 0.00284 0.00680 0.417 0.67656
I(career_year^2) class3 0.22747 0.00000 Inf 0.00000 ***
Trajectory shapes:
      class      b1      b2      shape peak_or_trough_year
Class 1 (n=53) -0.0361 0.00292 U-shape (dip & rise)          6.2
Class 2 (n=55) 0.0805 0.00284 U-shape (dip & rise)        -14.2
Class 3 (n=6) -2.3610 0.22747 U-shape (dip & rise)          5.2
Trajectories saved: rawquad_3class_trajectories.pdf

```

```

=====
Done. Check rawquad_fit_comparison.csv and BIC plot first.
Then open trajectory PDFs for each viable solution.
=====

```

```

There were 16 warnings (use warnings() to see them)
> source("rerun_lcga_b250.R")
=====

```

```

LCGA stability check: B = 250 random starts
=====

```

```

Faculty: 114 | Observations: 1774
Small-class threshold: 5 (5% of n)

```

```

--- Fitting 1-class anchor model ---
1-class BIC = 6010.6

```

```

--- sqrt(pub) linear, 4 classes, B = 250 ---
This may take several minutes ...
Done in 452.0 seconds.
log-lik = -2233.2048 | BIC = 4523.2 | entropy = 0.888 | AvePP = 0.937
Class sizes: C1=25, C2=55, C3=6, C4=28

```

```

--- Fitting quadratic 1-class anchor ---
1-class quadratic BIC = 6013.7

```

```

--- sqrt(pub) quadratic, 4 classes, B = 250 ---
This may take several minutes ...
Done in 913.0 seconds.
log-lik = -2220.2281 | BIC = 4516.2 | entropy = 0.905 | AvePP = 0.949
Class sizes: C1=6, C2=55, C3=25, C4=28

```

```

=== Stability comparison ===
      spec    log_lik      BIC entropy ave_pp small_class
sqrt_linear_4c_B50 -2243.600 4523.200 0.8880 0.9370      None
sqrt_linear_4c_B250 -2233.205 4523.244 0.8881 0.9366      None
sqrt_quad_4c_B50      NA 4516.200 0.9050 0.9560      None
sqrt_quad_4c_B250 -2220.228 4516.235 0.9055 0.9494      None

```

```

Linear BIC change (B50 → B250): +0.0
Quadratic BIC change (B50 → B250): +0.0
Linear vs Quadratic BIC gap at B250: +7.0 (positive = linear higher/worse)

```

```

>> Linear model is STABLE: B=250 solution is effectively the same as B=50.

```

```

Assignment change report written to: b250_assignment_changes.txt
=====

```

All outputs saved:

b250_stability_comparison.csv
b250_lin4_assignments.csv
b250_quad4_assignments.csv
b250_assignment_changes.txt

```
> source("run_h7_promotion_timing.R")
```

H7: Promotion Timing by LCGA Trajectory Class

Long-format rows loaded : 1777

Faculty in class file : 114

Excluding 3 faculty flagged never_asst_prof = TRUE.

H7 analytic sample: 111 faculty (3 excluded as never-Asst-Prof).

Promoted faculty (used for KW test): 73 of 111 (65.8%)

Saved: h7_faculty_level_dataset.csv

--- Class-level descriptives ---

trajectory_ordered	trajectory_label	class_assignment	n_faculty	n_promoted
Very low-output	Very low-output	1	27	21
Low-to-moderate rising	Low-to-moderate rising	4	23	14
Moderate-to-high rising	Moderate-to-high rising	2	55	35
Small high-growth	Small high-growth	3	6	3
pct_promoted	mean_career_length	mean_pub_per_year	mean_yrs_to_promo	median_yrs_to_promo
77.8	17.4	0.26	7.6	7
60.9	17.4	2.59	8.1	7
63.6	14.4	5.89	7.6	7
50.0	9.2	17.24	6.7	6
sd_yrs_to_promo	iqr_yrs_to_promo			
1.83	3.0			
3.48	3.8			
2.06	1.0			
1.15	1.0			

Saved: h7_class_descriptives.csv

FLAG: Very low-output class has the highest promotion rate (%.1f%).

77.8 This is counter to the simple H7 expectation.

--- Kruskal-Wallis test: years_to_promotion ~ trajectory_ordered ---

N (promoted faculty) = 73

Kruskal-Wallis H(3) = 0.908, p = 0.8234

Epsilon-squared = 0.0000 (small $\approx$.01, medium $\approx$.06, large $\approx$.14)

--- Pairwise Wilcoxon tests (BH correction) ---

Pairwise p-values (BH-adjusted):

group1	group2	p_adj_BH	significant
Low-to-moderate rising	Very low-output	0.9592329	FALSE
Moderate-to-high rising	Very low-output	0.9592329	FALSE
Small high-growth	Very low-output	0.9592329	FALSE
Moderate-to-high rising	Low-to-moderate rising	0.9592329	FALSE
Small high-growth	Low-to-moderate rising	0.9592329	FALSE
Small high-growth	Moderate-to-high rising	0.9592329	FALSE

Saved: h7_pairwise_wilcox.csv

Saved: h7_kruskal_results.txt

--- Survival analysis (all eligible faculty; non-promoted = censored) ---
N total = 111 | promoted = 73 | censored = 38

Cox PH model summary:

Call:

```
coxph(formula = Surv(time_to_event, event_promoted) ~ trajectory_factor,  
      data = h7_data)
```

n= 111, number of events= 73

	coef	exp(coef)	se(coef)	z	Pr(> z)
trajectory_factorLow-to-moderate rising	-0.43057	0.65014	0.35600	-1.209	0.226
trajectory_factorModerate-to-high rising	-0.05834	0.94333	0.27840	-0.210	0.834
trajectory_factorSmall high-growth	0.64854	1.91275	0.62638	1.035	0.300

	exp(coef)	exp(-coef)	lower .95	upper .95
trajectory_factorLow-to-moderate rising	0.6501	1.5381	0.3236	1.306
trajectory_factorModerate-to-high rising	0.9433	1.0601	0.5466	1.628
trajectory_factorSmall high-growth	1.9127	0.5228	0.5604	6.529

Concordance= 0.534 (se = 0.047)

Likelihood ratio test= 3.06 on 3 df, p=0.4

Wald test = 3.2 on 3 df, p=0.4

Score (logrank) test = 3.31 on 3 df, p=0.3

Log-rank test:

Call:

```
survdifff(formula = Surv(time_to_event, event_promoted) ~ trajectory_factor,  
          data = h7_data)
```

	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V
trajectory_factor=Very low-output	27	21	19.13	0.18237	0.3614
trajectory_factor=Low-to-moderate rising	23	14	17.75	0.79361	1.6234
trajectory_factor=Moderate-to-high rising	55	35	34.48	0.00779	0.0217
trajectory_factor=Small high-growth	6	3	1.63	1.14544	1.5122

Chisq= 3 on 3 degrees of freedom, p= 0.4

Proportional hazards assumption (cox.zph):

	chisq	df	p
trajectory_factor	1.84	3	0.61
GLOBAL	1.84	3	0.61

Saved: h7_cox_model_results.csv

Saved: h7_survival_diagnostics.txt

Saved: h7_boxplot_years_to_promotion.pdf

Saved: h7_km_plot.pdf

Saved: h7_results_summary.txt

```
=====
All H7 output files saved.
=====
```

```
>
```